



POWER API FILE FORMAT

V1.5.2

Version	Date	Comment
V 1.5	15/09/2021	International document creation
V 1.5.1	03/03/2022	Minor fix on unit
V 1.5.2	31/03/2022	Add some sensors

1 Introduction

PowerAPI file format allows to send hardware's data to Energysoft from an automaton or any other computer device.

2 Communication protocol

Two communication protocols are available: FTP and HTTP.

2.1 FTP protocol

In the web application, adding a PowerAPI type communication gateway to use this feature. Data should be sent to the gateway's name sub directory.

Data is sent in a CSV file (.csv extension) which can be compressed to GZIP format (.gz extension).

Be careful, data files are automatically deleted after integration in the software. As the treatment of received files is done on the fly, it is therefore expected not to find in the directory files which were just dropped.

FTP APPEND mode is not supported, it is necessary to send back a complete file upon every sending.

Recommendations

The file name must include the site name and the creation date of the file (free format). Sending several times data of the same date/time should be avoided. It is allowed for modification of previously sent data, but the sending mechanism must avoid sending the same data repeatedly.

We reserve the right to reject the site data in case of an inappropriate operation.

Please note that the FTP server includes low level security mechanism to avoid service degradation due to server overload. In particular, the FTP server may refuse a file if too many previous files are still pending processing for the site. The FTP client must be able to resend a file in the event of a file refusal or a temporary unavailability of the service.

2.2 HTTP protocol

Sending a CSV file is allowed as a Web Service call. This function is activated by default for any site declared on the portal.

Call

url : <https://energysoft.app/datapi/v2/data/post/<site>>

The call method is POST

Authentication

Authentication is done using the "Basic authentication" mechanism. The username and password provided by this mechanism must correspond to a valid API role user. The modifiable data will be restricted according to the rights granted to this user. To perform basic authentication add an "Authorization" header to the HTTP request according to Web standards.

Parameter

- site = energysoft site's reference

Post content

The type of content sent must be of the "text/csv" type (http Content-Type header). The content sent is a csv file which respects the format described in the next chapter. The service checks the existence and the right to access the site, then sends the file to a file queue to be processed. The file is therefore not processed during the execution of the Webservice but immediately afterwards. The processing time depends on the number of files awaiting processing for the site.

Output

If successful, the request returns an HTTP 200 code. If the site does not exist, the request returns an HTTP 404 code

Post example using curl

```
curl -i -user sampleuser:samplepassword -i -X POST
```

```
https://energysoft.app/datapi/v2/data/post/ref -H "Content-Type:text/csv" --data-
```

```
binary "@data.csv"
```

where

- ref : energysoft site's reference
- sampleuser : the user login
- samplepassword : the user password
- data.csv : the data file to be processed

3 CSV file format

The Power API format is a file format designed by S4E to insert measurements of all kinds into the Energysoft database. Two formats are available Simple CSV and Compact CSV. In both cases, they are files encoded in UTF-8 (UNICODE character set). The separator is the semicolon. The end of the line is \n or \r\n.

Regarding numbers: the decimal separator is the period, there is no thousands separator. A text field can be framed by double quotes. If it contains a semicolon, it must be surrounded by double quotes. End of line characters are not supported inside a text field.

The first line of the file is the header line.

CSV files can be sent in archive formats each archive containing a single csv file.

The supported formats are zip and Gnu gzip (expected extensions .zip, .csv.gz)

3.1 Simple CSV format

All column naming should use lower case.

First column is the date.

the 3 mandatory columns

- date : date of the measurement in the site's local time zone using the following format yyyy-MM-dd HH:mm:ss or yyyy-MM-ddTHH:mm:ss
- device : hardware type (inverter, meter, pyranometer, thermometer, site...)
- serial : device serial number or unique identifier (must be unique for a type of Device), may be ignored for the device type "site".

Optional columns : the following columns are related to measure. The column header contains the name of the measure in lower case.

Example :

date;device;serial;energy;energy_tot;energy_in;irradiation

2017-04-04 10:00:00;inverter;inv1;51.0;32445.0;48.3

2017-04-04 10:00:00;inverter;inv2;49.0;6327.0;47.0

2017-04-04 10:00:00;solarimeter;pyrano1;;;954

2017-04-04 11:00:00;inverter;inv1;65.0;32510.0;63.0

Energysoft / Power API File Format

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2017-04-04 11:00:00;inverter;inv2;56.0;6383.0;54.0

2017-04-04 11:00:00;solarimeter;pyrano1;;;1012

3.2 Compact CSV format

All column naming should use lower case.

First column is the date :

La première colonne est la date : date of the measurement in the site's local time zone using the following format yyyy-MM-dd HH:mm:ss or yyyy-MM-ddTHH:mm:ss

The date is followed by a sequence of n columns defining the measurements relating to a device.

The sequences are repeated as many times as device. Each sequence begins with a column named for the type of device (inverter, meter, thermometer, ...), this column will contain the serial number of the device. The next columns contain the measurements for this device.

Sequence example :

inverter;temperature;current_in;power_in;energy_tot;

XP1000-BP;136;0;0;22488;

Example of a complete structure :

date;inverter;temperature;current_in;power_in;energy_tot;volt_in;cosphi;frequency;reactive_power;power;current;current1;current2;solarimeter;irradiation;solarimeter;irradiation

2017-04-06T07:05:06;XP1000-BP;136;0;0;22488;6815;0;500;0;0;23;6;6;1;700;2;800

2017-04-06T07:15:06;XP1000-BP;138;0;0;22488;7440;-473;500;0;0;23;6;7;1;800;2;900

2017-04-06T07:25:06;XP1000-

BP;140;198;131;22498;6639;-2075;500;577;123;2975;1041;1020;1;932;2;1000

2017-04-06T07:35:07;XP1000-

BP;143;311;211;22528;6785;-3348;500;559;199;2972;1034;1048;1;933;2;1000

2017-04-06T07:45:07;XP1000-

BP;144;425;301;22563;7089;-4547;500;548;280;3056;1025;1100;1;928;2;1000

2017-04-06T07:55:07;XP1000-

BP;139;597;436;22623;7289;-6147;500;523;407;3270;1110;1147;1;924;2;1000

3.2 Reference tables

Measures tables

Device name	Measure name	Unit
meter	energy_tot	kWh
meter	energy	kWh
meter	power	W
meter	reactive_energy_tot	kVARh
meter	reactive_energy	kVARh
meter	negative_reactive_energy	kVARh
meter	positive_reactive_energy	kVARh
meter	tanphi	
meter	volt	V
meter	volt1	V
meter	volt2	V
meter	volt3	V
meter	current	A
meter	current1	A
meter	current2	A
meter	current3	A
meter	power	W
meter	power1	W
meter	power2	W
meter	power3	W
meter	frequency	Hz
meter	frequency1	Hz

meter	frequency2	Hz
meter	frequency3	Hz
meter	cosphi	
meter	cosphi1	
meter	cosphi2	
meter	cosphi3	
meter	apparent_power	VA
meter	apparent_power1	VA
meter	apparent_power2	VA
meter	apparent_power3	VA
meter	reactive_power	var
meter	reactive_power1	var
meter	reactive_power2	var
meter	reactive_power3	var
meter	riso	ohms
meter	energy_tot_rate_a	
meter	energy_tot_rate_b	
meter	energy_tot_rate_c	
meter	energy_tot_rate_d	
meter	energy_rate_a	
meter	energy_rate_b	
meter	energy_rate_c	
meter	energy_rate_d	
generic_meter	quantity	
generic_meter	index	
water_meter	volume	m3
water_meter	volume_tot	m3
gas_meter	volume	m3
gas_meter	volume_tot	m3
digital_io	status	
solarimeter	irradiation	W/m2
solarimeter	solar_energy	kWh/m2

solarimeter	global_horizontal_irradiation	W/m2
anemometer	wind_speed	Km/h
anemometer	wind_orientation	°
thermometer	temperature	°C
thermometer	module_temp	°C
hygrometer	hygrometry	%
weather_vane	wind_orientation	°
raingauge	rainfall	mm
flowmeter	flow	L/min
manometer	pressure	Pa
load_sensor	load	%
soiling_sensor	soiling_ratio	%
soiling_sensor	transmission_loss	%
inverter	energy_in	kWh
inverter	energy_tot_in	kWh
inverter	energy_tot	kWh
inverter	energy	kWh
inverter	operating_h_tot	h
inverter	feeding_operating_h_tot	h
inverter	nominal_power	Wc
inverter	temperature	°C
inverter	volt_in	V
inverter	current_in	A
inverter	power_in	W
inverter	volt	V
inverter	volt1	V
inverter	volt2	V
inverter	volt3	V
inverter	current	A
inverter	current1	A
inverter	current2	A

inverter	current3	A
inverter	power	W
inverter	power1	W
inverter	power2	W
inverter	power3	W
inverter	apparent_power	VA
inverter	apparent_power1	VA
inverter	apparent_power2	VA
inverter	apparent_power3	VA
inverter	reactive_power	var
inverter	reactive_power1	var
inverter	reactive_power2	var
inverter	reactive_power3	var
inverter	riso	ohms
inverter	frequency	Hz
inverter	frequency1	Hz
inverter	frequency2	Hz
inverter	frequency3	Hz
inverter	cosphi	
inverter	cosphi1	
inverter	cosphi2	
inverter	cosphi3	
inverter	voltL1to2	V
inverter	voltL2to3	V
inverter	voltL3to1	V
inverter	volt.mppt.n	V
inverter	current.mppt.n	A
inverter	power.mppt.n	W
inverter	volt.inverter_entry.n	V
inverter	current.inverter_entry.n	A
inverter	power.inverter_entry.n	W
junction	temperature	°C

junction	volt	V
junction	current	A
junction	power	W
junction	energy	kWh
junction	energy_tot	kWh
junction	volt.string.n	V
junction	current.string.n	A
junction	power.string.n	W
tracker	tilt	°
tracker	tilt_theoric	°
tracker	tilt2	°
tracker	tilt2_theoric	°
tracker	volt	V
tracker	current	A
tracker	power	W
battery	volt	V
battery	current	A
battery	power	W
battery	energy	kWh
battery	state_of_charge	%
battery	state_of_health	%
battery	nb_of_discharge_cycles	
battery	depth_of_discharge	%
battery	capacity	Ah
charger	volt	V
charger	current	A
charger	power	W
charger	energy	kWh
charger	energy_tot	kWh
charger	volt_in	V
charger	current_in	A
charger	power_in	W

charger	energy_in	kWh
charger	energy_tot_in	kWh
charger	volt1	V
charger	volt2	V
charger	volt3	V
charger	current1	A
charger	current2	A
charger	current3	A
charger	power1	W
charger	power2	W
charger	power3	W
wind_turbine	wind_orientation	°
wind_turbine	energy_tot	kWh
wind_turbine	energy	kWh
wind_turbine	power	W
wind_turbine	volt	V
wind_turbine	current	A
wind_turbine	nominal_power	W
wind_turbine	cut_in_wind_speed	m/s
wind_turbine	cut_out_wind_speed	m/s
wind_turbine	rated_wind_speed	m/s

* adding a "status" type measure will automatically create an entry in the status type events table (with name event_status), cf. Table of events.

Bidirectional meters

In EnergySoft, it is the meter typing that indicates whether the measurement refers to a production or consumption meter. In the case of bidirectional meters, standard measurements refer to production data. To transmit consumption data, add the prefix **in.** to standard measurements.

Meter type	Device name	Measure name	Unit
Production	meter	power	W
Consumption	meter	power	W
Bidirectional production	meter	power	W
Bidirectional consumption	meter	in.power	W

Custom measures

In addition to the above standard measures which are normalized (calculations, standard graphs), it is possible to add customized measures which will be used in personalized graphs. The measures must be integer or decimal numbers. This feature allows the use in Energysoft of measurements calculated by a third-party system (EMS, SCADA, PLC). The measurement can be attached to a device , if it exists, or to a site.

For example, adding two custom measures:

- expected charge of a battery
- oil pressure of a hydraulic circuit.

Device name	Measure name
meter	forecast_state_of_charge
site	oil_pressure

For the data to be integrated into Energysoft, you must create a custom variable of the type "Received measurement" which will define the name of the measurement used in the file (see [Add a custom variable](#))

Events table

Device name	Event name	Comment
inverter	error	hardware manufacturer error code
inverter	status	hardware manufacturer status code
inverter	sunspec.error	sunspec error code
inverter	sunspec.status	sunspec error status
inverter	error_<manufacturer error code>	1 / 0 ou true / false
inverter	status_<status code>	1 / 0 ou true / false
inverter	sunspec.error_<sunspec error code>	1 / 0 ou true / false
inverter	sunspec.status_<sunspec status code>	1 / 0 ou true / false
<any device>	event_xxxx	where xxxx is the event name
<any device>	error	manufacturer error code

Example :

date;device;serial;status

2017-04-04 10:00:00;digital_io;relay1;0

2017-04-04 10:00:00;digital_io;relay2;1

2017-04-04 10:10:00;digital_io;relay1;0

2017-04-04 11:00:00;digital_io;relay2;1